

Feature selection using Principal Component Analysis for massive retweet detection

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Abstract

Social Networks become a major actor in massive information propagation. In the context of the Twitter platform, its popularity is due in part to the capability of relaying messages (*i.e.* *tweets*) posted by users. This particular mechanism, called *retweet*, allows users to massively share tweets they consider as potentially interesting for others. In this paper, we propose to study the behavior of tweets that have been massively retweeted in a short period of time. We first analyze specific tweet features through a Principal Component Analysis (PCA) to better understand the behavior of highly forwarded tweets as opposed to those retweeted only a few times. Finally, we propose to automatically detect the massively retweeted messages. The qualitative study is used to select the features allowing the best classification performance. We show that the selection of only the most correlated features, leads to the best classification accuracy (F-measure of 65.7%), with a gain of about 2.4 points in comparison to the use of the complete set of features.

Keywords:

Massive retweet, Principal Component Analysis, Feature selection, Classification

1. Introduction

Twitter is one of the most popular microblogging services which allows real-time traffic, publication, and sharing of user opinions or stories (Pak and Paroubek, 2010). It also has an impact on economics and politicals (Bollen et al., 2011).

Text messages sent through the Twitter platform, called *tweets*, are composed of up to 140 characters. The power of this microblogging service lies in the fact that users can relay or forward a tweet using the *retweet* paradigm. Thus, information posted by a user can have a very different impact on his environment depending on the number of times it is relayed (*i.e.* retweeted) over a short period of time. For this reason, studying the massive retweet behavior is crucial in information propagation.

Few studies have highlighted the influence of tweet features in the massive information propagation. In this paper, we present a work that aims at detecting the fast and massive spread of short text messages from the Twitter service. We propose to analyze a set of features related to the user environment and to the message content. This study evaluates the impact of each feature in the retweet phenomenon. Finally, two massive retweet classification approaches are compared using the most correlated retweet features chosen from the qualitative study.

The next section enumerates related works about information propagation on social networking services. The experimental protocol for the massive retweet analysis is in section 3. Then, section 4 highlights the correlation between tweet features and the massive retweet phenomenon, while section 5 presents experiments to evaluate the effectiveness of this analysis using two different classification methods. Finally, conclusions and perspectives are given in section 6.

2. Related work

Although Twitter is a recent information-sharing model, its service mechanisms have been widely studied. In (Zhao and Rosson, 2009; Kwak et al., 2010), the authors analyze Twitter in a general case and study its social impact. Other works have focused on various aspects of Twitter, such as event detection (Phelan et al., 2009; Zhao et al., 2011), user influence (Cha et al., 2010), sentiment analysis (Tumasjan et al., 2010), prevention of climate disaster (Sakaki et al., 2010), or hashtag analysis (Huang et al., 2010).

The particular retweet mechanism has been studied in order to predict the number of retweets in a particular range. In (Suh et al., 2010), the authors proposed to extract a set of features to explain this mechanism. The prediction of potential popular messages was proposed by (Zaman et al., 2010; Hong et al., 2011).

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The behavior of Twitter users is also studied. In (Luo et al., 2013), the authors analyze the retweet behavior and predict who will retweet a given tweet. Features that influence the retweet probability, such as information freshness, user sending rate or tweet size, have been analyzed by (Comarella et al., 2012). In (Macskassy and Michelson, 2011), the authors analyze the retweet information diffusion while the work by (Yang and Counts, 2010) evaluates the impact of the network structure on this phenomenon.

3. Experimental protocol

The study of the massive retweet phenomenon requires a set of relevant features and a large experimental corpus. Table 1 presents the most used features (Suh et al., 2010; Luo et al., 2013) for our preliminary qualitative study of the massive retweet mechanism, including tweet content and user features.

Table 1: Description of tweet features for a user U .

Content features	Description	Twitter API names
retweet	# of sharing	<i>retweet.count</i>
hashtag	# of topics in a tweet	<i>hashtags</i>
mention	# of cited usernames	<i>text</i>
url	# of contained URLs	<i>urls</i>
User features	Description	Twitter API names
days	# of days U created its account	<i>U/created_at</i>
favorite	# of favorite tweets by U	<i>U/favourites.count</i>
follower	# of users who follow U	<i>U/followers.count</i>
followee	# of friends of U	<i>U/friends.count</i>
status	# of tweets wrote by U	<i>U/statuses.count</i>

A corpus of 6 million tweets was collected using the official Twitter API¹. The emission dates of the tweets were from April 14th 2006 (d_{start}) to May 13th of 2011 (d_{end}) (lifetime period of 265 weeks which corresponds to about 5 years). Out of this set, 30,903 tweets were retweeted at least once and up to more than 100 times (100+). This tweet set is used to evaluate the correlation between the features described above. Note that, at the time of the corpus extraction, the Twitter service did not allow us to have the exact retweet number for tweets retweeted more than 100 times.

Figure 1 presents the tweet distribution depending on the number of retweets. We can see that tweets are mostly retweeted either a few times (43% for less than 30) or massively (44% for more than 100 times), while only 13% are retweeted between 30 and 100 times.

A natural intuition about the retweet behavior would be to consider that older tweets would be more likely to be massively retweeted than recent ones. Our assumption is that the age of a tweet has a limited impact on its number of retweets since information in social media has a tendency to be either massively spread in a short time period or ignored (Morchid et al., 2014). To evaluate the impact of a tweet lifespan in the retweet behavior, let's define

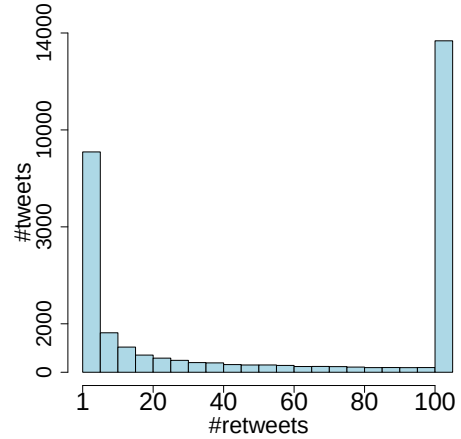


Figure 1: Tweet distribution depending on the number of retweets.

γ as the proportion of seldom ($\gamma_{seldom}(d)$) or massively ($\gamma_{massive}(d)$) retweeted tweets knowing the broadcasting date ($d_{created}$) of the tweet t , and δ the difference:

$$\delta(d) = |\gamma_{seldom}(d) - \gamma_{massive}(d)|, \quad (1)$$

where d is the number of weeks (*i.e.* a lifetime period) between $d_{created}$ and d_{end} ($d_{created} \geq d_{end}$). We chose to consider as *seldom* a tweet retweeted less than 30 times and *massive* a tweet retweeted more than 100 times in order to maintain a comparable dataset size. A straightforward study of the differences between seldom and massively retweeted messages shows that the mean of δ is 0.204% and its standard deviation is 0.206%, which confirms our assumption: the age of a tweet does not influence its low or massive spreading. Figure 2-(a) presents the proportion of $\gamma_{seldom}(d)$ and $\gamma_{massive}(d)$ tweets for a lifetime of less than 20 weeks before d_{end} , while Figure 2-(b) presents exactly the same curves but with a log scale for a lifetime d greater than 20 weeks. Figure 2-(b) allows us to point out that difference between the proportion of seldom or massively retweeted tweets follows the same curve even if the proportion of this difference (δ) is small (less than 1%). Moreover, the main proportion of tweets in both subsets, is more or less concentrated during the same epoch (see Figure 2-(a) $0 \leq d \leq 2$).

4. Analysis of Massive retweet factors

In this section, we propose to use this tweet dataset to study the correlation between retweet features described in table 1. A Principal Component Analysis (PCA) is applied to transform these features from an original space representation of correlated variables, into a set of linearly uncorrelated variables called *principal components* (Factors). Carrying out PCA on the features, including retweet feature, allows to sort and group these variables with respect to the total variability and to assess how the retweet variable affects this variability.

¹<http://dev.twitter.com>

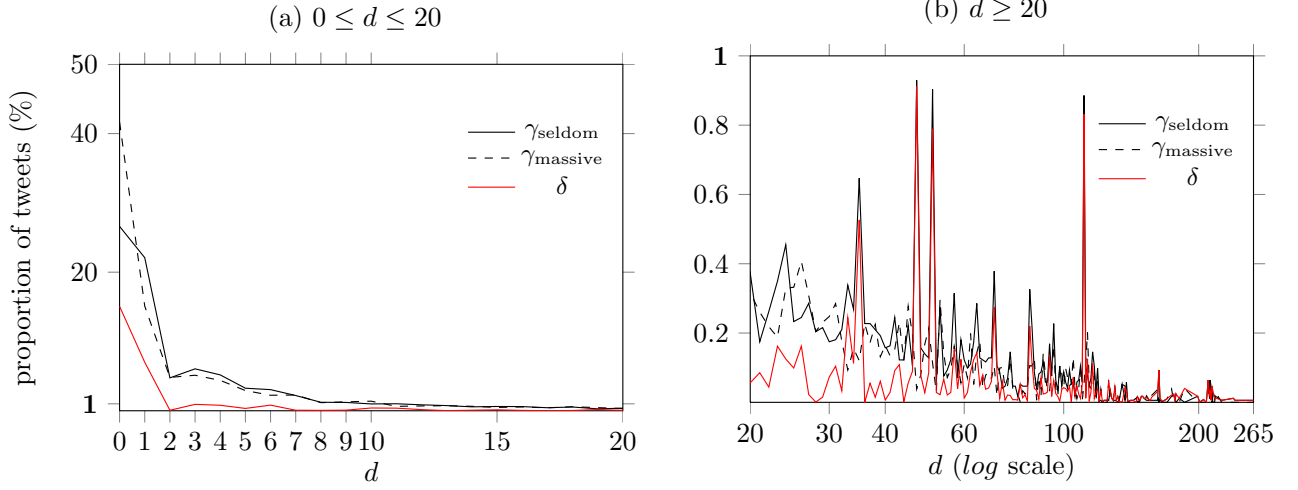


Figure 2: Proportion of seldom or massively retweeted tweets during their lifetime d ($d = d_{created} - d_{end}$) up to 20 weeks (a) and between 20 and 265 weeks (b).

4.1. Principal Component Analysis

The Principal Component Analysis (PCA) is used by almost all scientific disciplines and is probably the most popular multivariate statistical technique. Introduced for the first time by (Cauchy, 1829; Pearson, 1901), its more recent instantiation was formalized by (Hotelling, 1933) who also introduced the term *principal component*. The goals of PCA are to:

- extract the most important information from the data table
- compress the size of the dataset by keeping only this important information
- explain and simplify the description of the dataset
- analyze the structure of observations and variables

PCA computes new variables called *principal components* to achieve these goals. These variables are obtained as linear combinations of the original variables. The first principal component is required to have the largest possible variance to “explain” the largest part of the inertia of the dataset. Then, the second component is computed under the constraint of being orthogonal to the first component and to have the largest possible inertia. The other components are computed in the same way. The values of these new variables are called *factor scores* and are interpreted geometrically as the *projections* of the observations onto the principal components.

These are obtained from the Singular Value Decomposition (SVD) of the dataset $\mathbf{X}$, with:

$$\mathbf{X} = \mathbf{P}\mathbf{\Delta}\mathbf{Q}^T \quad (2)$$

where $\mathbf{P}$ is a $m \times l$ matrix of left singular vectors, $\mathbf{Q}$ is the $n \times l$ matrix of right singular vectors and $\mathbf{\Delta}$ is

the diagonal matrix of singular values. l is the rank of the matrix $\mathbf{X}$ ($l \leq \min\{m, n\}$). The $m \times n$ matrix of factor scores denoted $\mathbf{F}$ is obtained as $\mathbf{F} = \mathbf{P}\mathbf{\Delta}$ and can be interpreted as a *projection* matrix because multiplying $\mathbf{X}$ by $\mathbf{Q}$ gives the values of the *projections* of the observations on the principal components using equation 2:

$$\mathbf{X}\mathbf{Q} = \mathbf{P}\mathbf{\Delta}\mathbf{Q}^T\mathbf{Q} = \mathbf{P}\mathbf{\Delta} = \mathbf{F} \quad (3)$$

More information about PCA and SVD can respectively be found in (Abdi and Williams, 2010) and in (Golub and Reinsch, 1970).

4.2. Factor selection

As described in equation 3, matrix $\mathbf{F}$ is generated using a standardized dataset matrix $\mathbf{X}$ with $\dim(\mathbf{X}) = (m, n)$, where m is the number of observations ($m = 30,903$) and n ($n = 9$) is the number of variables or features. A PCA is then performed to find a subspace of size p , where the features are clustered depending on their projections into the factor space.

A set of p factors ($p \leq n$) is then selected. Each factor represents a certain part of the total variance of the dataset. Table 2 and figure 3 present the factors and their associated eigenvalues. Columns 3 and 4 of Table 2 respectively represent the percentage of the total variance and the cumulative variance accounted for from the original dataset. These factors are sorted in descending order depending on their eigenvalues.

4.3. Feature correlation

Several methods are used to estimate the *correct* number of factors to represent the data variance and the feature correlation. One of the most popular is the Kaiser Criterion (Kaiser, 1961). This rule recommends retaining factors with eigenvalues equal to or greater than 1. Moreover,

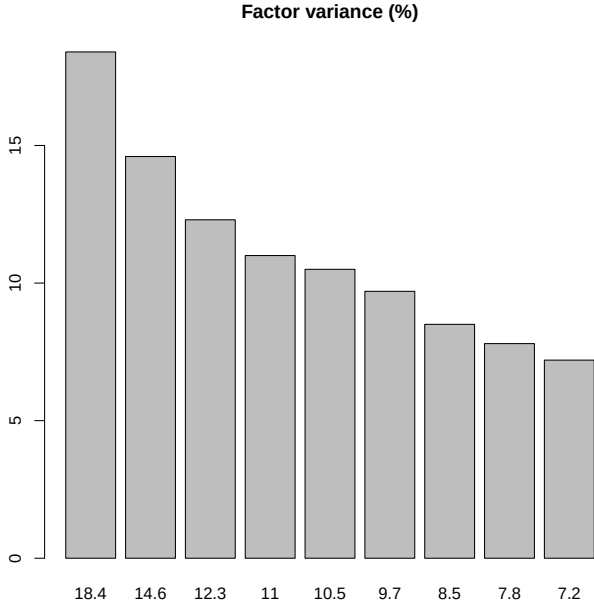


Figure 3: Proportion of the total variance contribution for each of the 9 studied features.

Table 2: Principal components from a PCA.

Component (Factor f_i)	Eigenvalues ev_i	Eigenvalues % variance $v(f_i)$	Cumulative % variance
1	1.65	18.41	18.41
2	1.31	14.57	32.98
3	1.11	12.33	45.31
4	1.00	11.02($\simeq \frac{1}{9}$)	56.34
5	0.94	10.50	66.84
6	0.87	9.71	76.55
7	0.76	8.47	85.02
8	0.70	7.79	92.82
9	0.64	7.17	100

in the original space representation, each of the 9 features contains about 11% ($\frac{1}{9}$) of the total variance explained by all the features. The principal component retained should then explain at least 11% of the total variance. The variance of a factor f_i is performed as follows:

$$v(f_i) = \frac{ev_i}{\sum_{j=0}^n ev_j} \quad (4)$$

As a result, the application of these two rules allows the selection of only the first 4 factors (see Table 2).

4.4. Orthogonal rotation

in order to facilitate the interpretation after determining the number of components, analysis often involves a rotation of the components that were retained. Two types of rotation are mainly used: *orthogonal* when the new axes are also orthogonal to each other, and *oblique* when the

new axes are not required to be orthogonal. However, the part of the inertia explained by the total subspace after rotation is the same as it was before the rotation (only the partition of the inertia has changed). In this paper, the orthogonal rotation (VARIMAX) is chosen to perform a *transformation* of the original data. VARIMAX rotation, developed by (Kaiser, 1958), is the most popular rotation method. This method assumes that a simple solution means that each component has a small number of large loadings, and a large number of zero loadings. Formally, VARIMAX searches for a linear combination of the original factors such that the variance of the squared loadings is maximized, which amounts to maximizing v :

$$v = \sum (q_{j,l}^2 - \bar{q}_l^2)^2 \quad (5)$$

with $q_{j,l}^2$ being the squared loading of the j^{th} variable of the matrix $\mathbf{Q}$ on the component l and $\bar{q}_l^2$ being the mean of the squared loadings.

This rotation is performed using the diagonal matrix of singular values and the eigenvectors associated with the correlated matrix of $\mathbf{X}$.

4.5. Factor mapping

These factor loadings almost always refer to the elements of matrix $\mathbf{Q}$. Table 3 presents factor loadings from matrix $\mathbf{F}$ for each feature in the subspace represented by the p first factors ($p = 4$) of Table 2. Each factor is independent and describes the correlation between the features in the original space representation.

Table 3: Feature projection in the factor space.

Content features	Factor 1	Factor 2	Factor 3	Factor 4
retweet	-0.3295	0.7009	-0.0072	0.1481
hashtag	0.0112	-0.1757	-0.2089	0.9306
mention	0.2569	-0.5927	-0.1982	-0.2175
url	0.1682	-0.5240	0.1599	0.1431
User features	Factor 1	Factor 2	Factor 3	Factor 4
days	0.5071	0.0901	0.4011	0.1717
favorite	0.5389	0.1974	0.3391	-0.0007
follower	0.3745	0.1678	-0.6694	-0.0795
followee	0.5133	0.2025	-0.4886	-0.0062
status	0.7092	0.2178	0.1964	0.0232

Figure 4-A shows that Factor 2 distinguishes the content (*url*, *mention*) from the user account features (*follower*, *followee*, ...). Tables 2 shows that Factor 2 carries more than 14% of the total variability of the dataset, and Table 3 shows that this variability is explained by the antagonism between the retweet feature (+0.7009) on one hand, and the *url* (−0.5240) and *mention* (−0.5927) features on the other hand. Moreover, 18% of the variability ($v(f_1)$, see Table 2) is carried by a strong correlation of the user features.

The *hashtag* feature is mainly carried by Factor 4 (+0.9306 see table 3) and explains almost 11% of the total variability (see table 2). Factor 3, which represents about 12% (see Table 2) of the variability, is represented by the

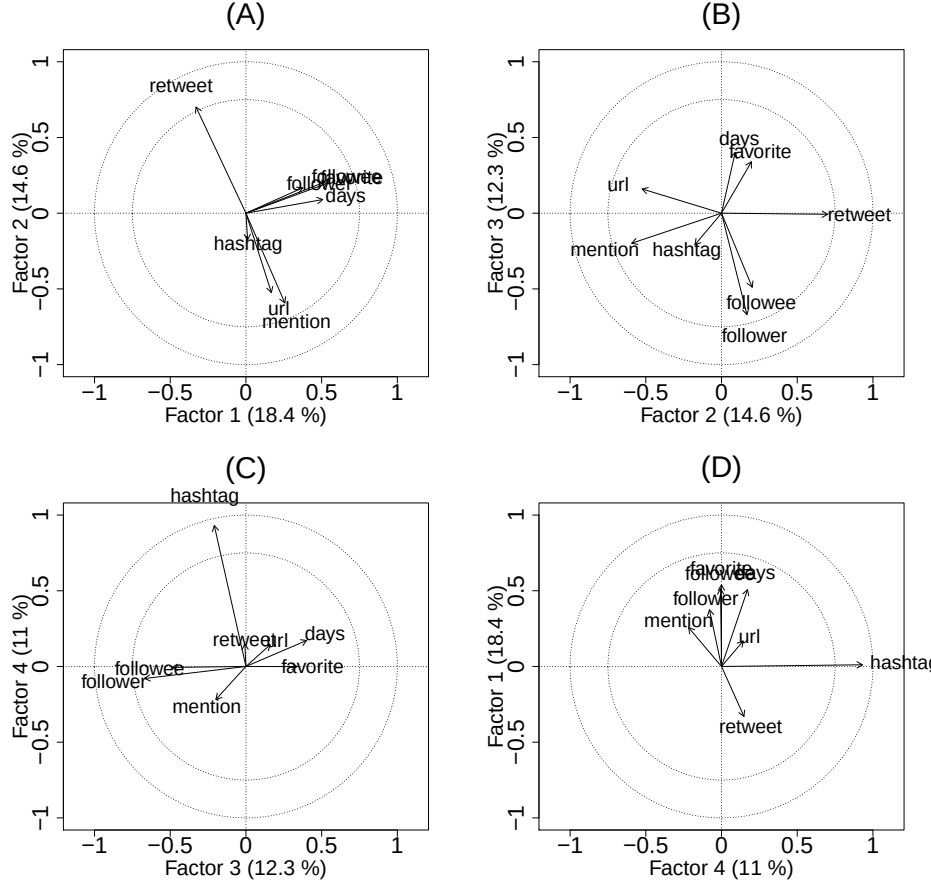


Figure 4: PCA factor map of the tweet features studied.

follower feature as shown in Figure 4-B and C. Then, this feature could be considered as partially independent. Figure 4 and Table 3 highlight the fact that most of the user profile features (*days*, *favorite*, *followee* and *status*) draw a unique feature cluster in the reduced subspace of factors.

4.6. Ordering features by relevance

The previous factor analysis allows us to sort features according to their impact on the total variability and their correlation with the retweet feature: *hashtag* (93% of Factor 4 total variability), *follower* (explains the main variability of Factor 3), *mention* (shares the variability with the retweet feature and explains most of the variability of Factor 2), *status* (explains more than 70% of Factor 1 variability but it shares this variability with 3 other features, as shown in Figure 4-A), and *url* (a feature that completes the retweet factor variance and shares this factor variability with *retweet* and *mention* features). While these 5 major features describe the main part of the total variance of the dataset, the last 3 features represent the residual variance of Factor 1: *favorite*, *followee* and *days*.

In the next section, these sorted features are used to automatically detect if a tweet will be massively retweeted.

5. Massive retweet detection

A system of massive retweet detection is created to evaluate the effectiveness of the ordered set of features (see section 4.6). Two classes are considered: the low retweeted tweets (LRT) and the massively retweeted tweets (MRT). These two classes are collected in the same time range. Indeed, although a LRT could become a MRT in the future, the aim of the proposed approach is to study and detect the tweets that are massively retweeted, but not to predict the potential number of retweets during the whole duration of their lifetime. MRT is composed of tweets retweeted more than 100 times (44%). To maintain a comparable data size, the LRT set has been chosen from the tweets retweeted up to 30 times (43%). Note that the tweets retweeted between 30 and 100 times (13%) are not considered in this study (see figure 1).

In order to better choose a suitable classification approach, we propose to analyze the class separability first.

5.1. Class separability

The linear separability between two classes can be estimated with the variation proportion explained by their mean shifts (Fukunaga and Hostetler, 1975). This value is equal to 1.37%, which indicates that 98.63% of the total

variability is due to the within-class dispersion. Figure 5 displays the data projection on their two principal axes, which explain 33% of the total variability ($v(f_1) + v(f_2)$, see Table 2), and the density ellipses of the two classes. It clearly indicates an inability to linearly separate these two classes.

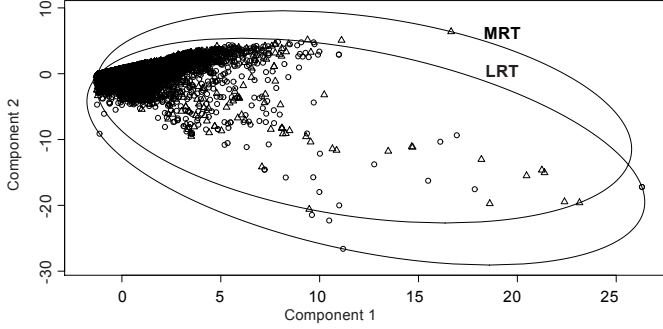


Figure 5: Data projection and density ellipses.

5.2. Experiments

Since we proved that the features are not linearly separable, we propose to use an efficient non-linear classification approach: the Support Vector Machines (SVM) (Vapnik and Lerner, 1963). Nonetheless, we can see in Figure 1 two distinct distributions for the LRT and MRT classes. This might indicate a power-law distribution that a parametric statistical analysis could capture. For this reason, a Naive Bayes approach (Rish, 2001) is also performed for comparison purpose with the SVM classification method.

An SVM and a Naive Bayes classifier are then trained with the vector representation of features to automatically detect if a tweet is massively retweeted or not. For the SVM approach (Yuan et al., 2012) with a polynomial kernel is performed. The Naive Bayes method is performed using the WEKA data mining software (Hall et al., 2009). Each class (LRT and MRT) has a training set of 12,195 tweets and a test of 1,355 tweets.

Tables 4 and 5 present the global classification results obtained on the LRT and MRT classes in terms of recall, precision, F1-measure, and false positive (FP) rate, respectively with the Naive Bayes and the SVM classification approaches. The F-measure is the harmonic mean of precision and recall and traditionally named *balanced F-score* as well. These four measures are defined as:

$$\text{Precision} = \frac{\# \text{tweets classified into class } i \text{ (LRT or MRT)}}{\# \text{tweets classified as class } i \text{ (LRT or MRT)}}$$

$$\text{Recall} = \frac{\# \text{tweets classified into class } i \text{ (LRT or MRT)}}{\# \text{tweets belonging to the class } i \text{ (LRT or MRT)}}$$

$$\text{F1-measure} = 2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

and

$$\text{False Positive rate} = \frac{\# \text{FP} + \# \text{FN}}{\# \text{FP} + \# \text{FN} + \# \text{TP} + \# \text{TN}}$$

where FP, FN, TP and TN respectively correspond to False Positive, False Negative, True Positive and True Negative.

Moreover, the number of features used varies from 1 to 8. The first system configuration uses only the most relevant feature (*hashtag*), the second system configuration uses the two most relevant ones (*hashtag* + *follower*) and so on, following the order defined in section 4.6.

Table 4: Global classification performance and false positive (FP) rate with Naive Bayes approach.

# Features	Recall	Precision	F1	Impr. (%)	FP
1	44.5	40.9	42.6	–	55.6
2	52.7	54.3	53.5	+20.4	47.2
3	57.5	60.0	58.7	+8.9	42.5
4	58.2	60.1	59.1	+0.7	41.8
5	59.1	60.2	59.6	+0.8	40.8
6	58.6	60.0	59.3	-0.5	41.3
7	58.5	60.3	59.4	+0.2	41.5
8	58.2	59.6	58.9	-0.8	41.8

Table 5: Global classification performance and false positive (FP) rate with SVM.

# Features	Recall	Precision	F1	Impr. (%)	FP
1	50.0	24.9	33.3	–	50.1
2	57.4	58.0	57.7	+42.3	42.6
3	63.0	66.9	64.9	+11.1	36.9
4	62.7	66.8	64.7	–	37.2
5	64.1	67.8	65.9	+0.3	35.9
6	62.8	66.8	64.7	-1.9	37.1
7	62.2	65.8	63.9	-1.3	37.6
8	62.4	64.7	63.5	–	37.6

First, we can see that the best classification results are obtained using the 5 most pertinent massive retweet features, irrelevant of the classification approach used (Naive Bayes or SVM). This confirms the intuition that motivated this work: while many features are relevant to explain the retweet phenomenon, only some of them are useful to detect massive retweets. These 5 features are sufficient to describe the main variability of the dataset and their correlation. We can also note that results obtained with the SVM classification approach are clearly better than the ones obtained with the Naive Bayes method (see Tables 4 and 5). This also validates the class separability analysis presented in section 5.1.

In details, by analyzing results obtained with the SVM approach in Table 5, we observed that the performance of the massive retweet detection (MRT), with a recall of 86.9% and a precision of 59.8%, is much better than the performance of the low retweet detection (LRT), with a recall of 41.2% and a precision of 75.8% (5 features). This seems coherent since the proposed study focused on the features correlated with the massive retweet phenomenon. The lower LRT classification results could also be explained by the fact that this study is not a prediction task: a tweet having a low retweet number at instant t does not mean that it will not be massively retweeted in the future.

Thus, it seems more difficult to correctly classify the LRT class without knowing its evolution in time.

6. Conclusions and Perspectives

In this paper, we proposed a preliminary work to detect tweets massively retweeted through an analysis of selected features. We have shown that the choice of the most relevant features has a real impact in the massive retweet detection with a gain of about 2.4% with the SVM and 0.7% with the Naive Bayes methods on the F-measure using the most correlated features in comparison to the use of all the features.

In future research, we shall analyze the classification results obtained on the low retweet (LRT) class. As highlighted in our study, the emission date of the tweet could be the reason of its low number of retweets: at a later date, this tweet could be massively retweeted. Then, we propose to not only use the number of retweets, but to incorporate its emission date. We could also take into consideration the temporal evolution of the corpus by studying the retweet number at two distinct moments in time (Kupavskii et al., 2012). This differs from our initial study which aims at analyzing the MRT phenomenon over a short period of time, but it would be interesting to compare the behavior of features in these two MRT contexts. The proposed approach mixes user and tweet features. An interesting work would be to study either side separately. Finally, more retweet features will be studied, such as the influence of the user followers (Ye and Wu, 2010), the Twitter social network (Java et al., 2007) or its apparition context (Morchid et al., 2013).

In this work, we focused on the selection of features using the PCA method since it was already successfully applied in the context of social networks (Liu, 2007; Suh et al., 2010). An interesting extension would be to compare this approach with traditional supervised feature selection methods, such as those described in (Zhao and Liu, 2007; Song et al., 2007).

On the other hand, we think that our techniques may be useful to study tweet filtering, topic priority detection, and polarity for reputation (see the CLEF RepLab campaign²). In this case, several approaches using several machine learning methods must be employed. In particular, we believe that it is interesting to exploit tweet contents, the selected metadata or the integration of external information, such as Wikipedia and users' homepages.

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